DIJSTRAK ALGORITHM VISUALIZER

A PROJECT REPORT

Submitted by

Akshat Raj -24BCS11369

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INTRODUCTION

The Dijkstra Algorithm Visualizer is an interactive web-based application designed to demonstrate the working of the Dijkstra's Algorithm in a simple and visual manner. Dijkstra’s Algorithm is one of the most important algorithms in graph theory, widely used to find the shortest path between nodes in a weighted graph.

Understanding how this algorithm works can be difficult when studied only through theory. This project aims to bridge that gap by providing a dynamic visualization where users can observe each step of the algorithm in real time. It allows users to select source and destination nodes, view how distances are updated, and see how the shortest path is determined.

The visualizer highlights visited nodes, updates tentative distances, and gradually builds the shortest path, making it easier for students and beginners to grasp the logic behind the algorithm. By combining interactive UI elements with core Data Structures and Algorithms concepts

OBJECTIVES

The main objectives of this project are:

- To provide an interactive platform for understanding Dijkstra's Algorithm
- To visualize the step-by-step process of finding the shortest path in a weighted graph
- To help users understand how distances are calcula ted and updated dynamically
- To demonstrate the selection of the minimum distance node at each step (greedy approach)
- To allow users to interact with the graph by selecting source and destination nodes
- To enhance conceptual clarity of graph-related data structures and algorithms
- To bridge the gap between theoretical learning and practical implementation

METHODOLOGY

- The project is developed as a web-based application using HTML, CSS, and JavaScript to provide an interactive user interface.
- A graph structure is created consisting of nodes (vertices) and edges (connections), where each edge has an associated weight.
- The user selects a source node and optionally a destination node through the interface.
- All node distances are initially set to infinity, except the source node which is set to zero.
- The system applies Dijkstra's Algorithm to compute the shortest path.
- At each step, the algorithm selects the unvisited node with the minimum distance (greedy approach).
- The distances of neighboring nodes are updated if a shorter path is found through the current node.
- Visited nodes are marked visually, and their distance values are displayed dynamically.
- The process continues until all nodes are visited or the destination node is reached.
- The final shortest path is highlighted on the graph for better understanding.
- Animations and step-by-step transitions are used to help users clearly observe how the algorithm progresses.

RESULT

- The Dijkstra Algorithm Visualizer was successfully developed and tested as a web-based application.
- The system correctly computes the shortest path between the selected source and destination nodes using Dijkstra's Algorithm.
- The visualization clearly displays each step of the algorithm, including node selection, distance updates, and path formation.
- Users are able to interact with the interface by selecting nodes and observing real-time changes in the graph.
- The final shortest path is accurately highlighted, making it easy to understand the outcome of the algorithm.

CONCLUSION

The Dijkstra Algorithm Visualizer project was successfully designed and implemented to demonstrate the working of Dijkstra's Algorithm in an interactive and user-friendly manner. The project effectively converts theoretical concepts of graph algorithms into a visual experience, making it easier to understand the process of finding the shortest path between nodes.

Through this project, important concepts of Data Structures and Algorithms such as graphs, greedy techniques, and path optimization were practically implemented and explored. The step-by-step visualization helps users clearly observe how distances are updated and how the final shortest path is determined.

Overall, the project achieves its objective of enhancing learning through visualization and provides a useful educational tool for students. It also demonstrates how algorithmic concepts can be applied in real-world scenarios such as navigation systems and network routing